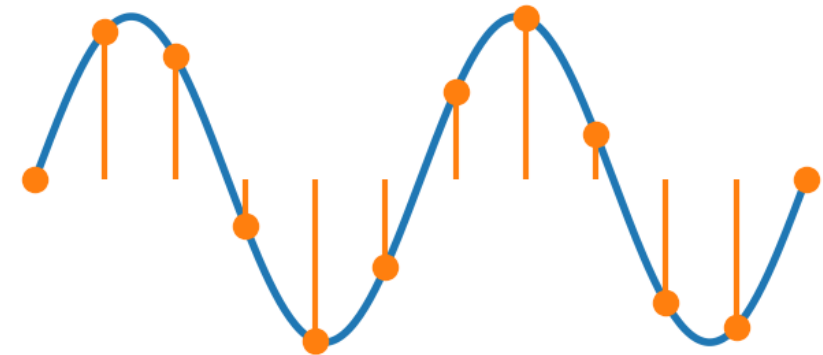


Topic 4

Sampling and Aliasing

From continuous-time signals to discrete-time sequences and reconstruction



Key idea. Sampling is safe only when the sampling rate is high enough to avoid aliasing.

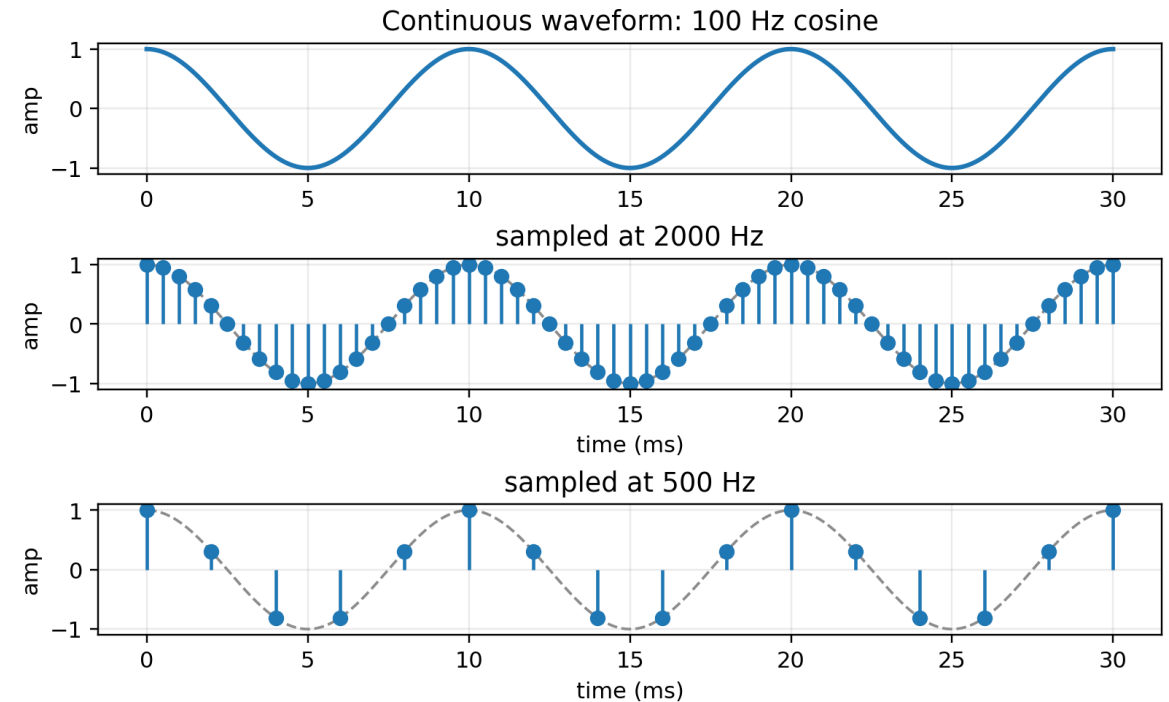
Sampling

From a continuous waveform to a sequence of numbers

$$x[n] = x(nT_s), \quad f_s = 1/T_s$$

T_s is the sampling period in seconds.
 f_s is the sampling rate in samples per second.

Higher sampling rates describe the waveform more densely.



Key idea. A sampled signal is not continuous in time; it is an indexed sequence.

Sampling sinusoidal signals

Continuous frequency becomes normalized discrete-time frequency

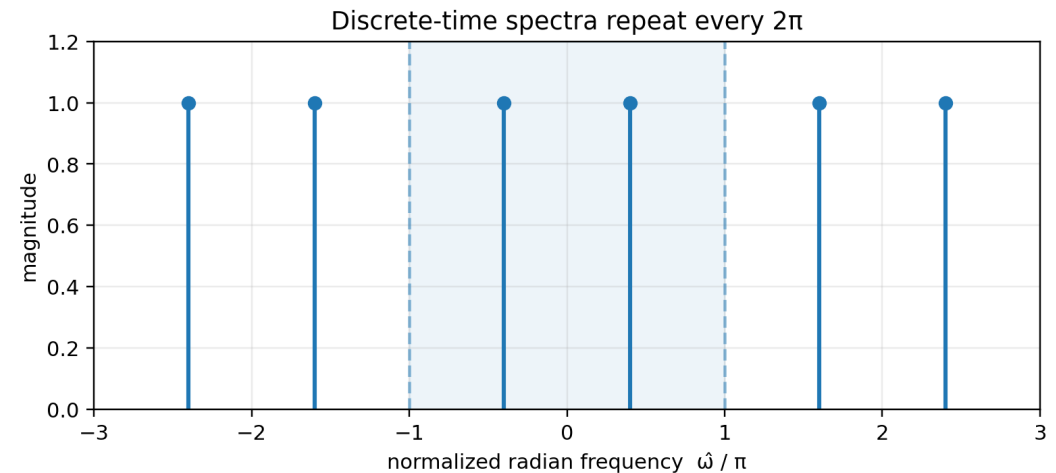
$$x(t) = A \cos(\omega_0 t + \varphi) \rightarrow \\ x[n] = A \cos(\hat{\omega}_0 n + \varphi)$$

$$\hat{\omega}_0 = \omega_0 T_s = 2\pi f_0 / f_s$$

ω_0 is in radians/second.

$\hat{\omega}_0$ is in radians/sample.

The sequence depends on frequency only modulo 2π .



Key idea. Discrete-time frequency is normalized by the sampling rate.

Example 1 — sampling a sinusoid

From analogue frequency to normalized frequency

Problem

$x(t) = 3\cos(2\pi \cdot 300t + \pi/6)$ is sampled at $f_s = 2000$ samples/s. Find T_s , $x[n]$, normalized frequency, and samples per period.

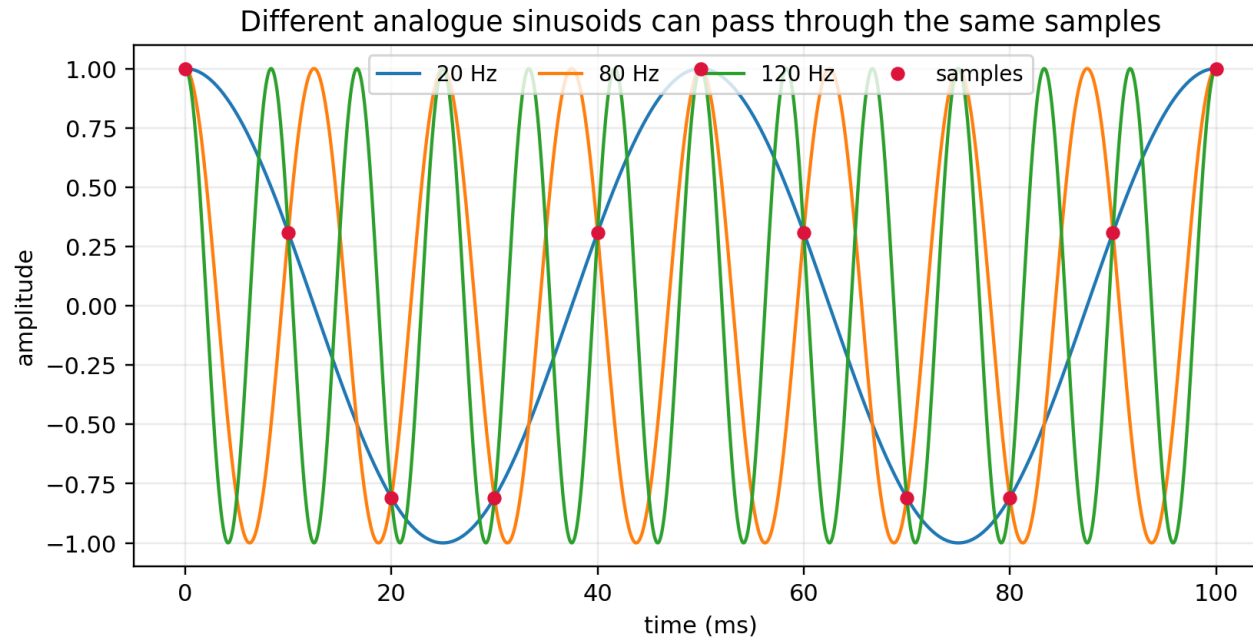
Solution

- 1 Sampling period: $T_s = 1/f_s = 1/2000 \text{ s} = 0.5 \text{ ms}$.
- 2 Normalized cyclic frequency: $\hat{f}_0 = f_0/f_s = 300/2000 = 0.15$ cycles/sample.
- 3 Normalized radian frequency: $\hat{\omega}_0 = 2\pi\hat{f}_0 = 0.3\pi$ rad/sample.
- 4 Samples per period: $f_s/f_0 = 2000/300 \approx 6.67$ samples.

Final answer: $x[n] = 3\cos(0.3\pi n + \pi/6)$, with $T_s = 0.5 \text{ ms}$ and $\hat{f}_0 = 0.15$ cycles/sample.

Aliasing

Different analogue frequencies can give the same samples



f and $f \pm mf_s$ are aliases

f and $mf_s - f$ are folded aliases

- The samples alone do not reveal the original analogue frequency.
- A band-limit assumption is needed for unique reconstruction.

Key idea. Aliasing means two different continuous-time signals have the same discrete-time name.

Alias families

Frequency repetition and folding

$$\hat{\omega} \equiv \omega + 2\pi m$$

$$\cos(\omega n + \varphi) = \cos((- \omega)n - \varphi)$$

$$f_{\text{alias}} = |f_0 - mf_s|$$

- Adding 2π to normalized frequency does not change the sequence.
- For cosines, negative frequency can appear as a phase-reversed folded alias.
- The observable frequency is usually taken inside the Nyquist interval.

Key idea. Aliasing is both a time-domain identity and a spectral periodicity.

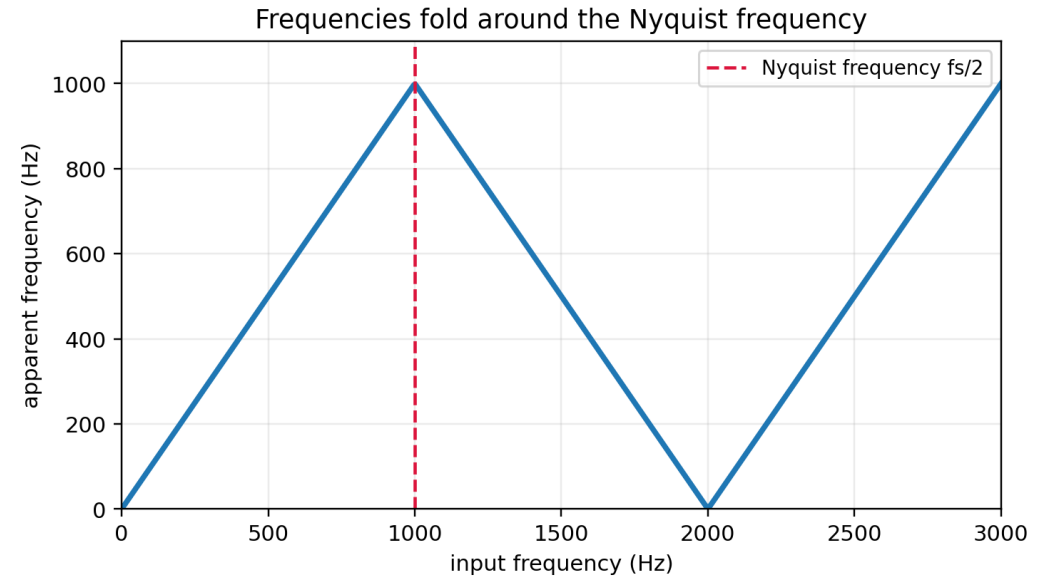
The sampling theorem

When exact reconstruction is possible

$$f_s > 2f_{\max}$$

$$f_N = f_s / 2$$

- f_N is the Nyquist frequency.
- No signal component may exceed f_N .
- CD audio: 44.1 kHz is slightly above twice 20 kHz.



Key idea. Exact reconstruction requires sampling faster than twice the highest analogue frequency.

Example 2 — choosing a sampling rate

Using the sampling theorem with several frequency components

Problem

$$x(t) = 2 + 4\cos(2\pi \cdot 400t) + 3\sin(2\pi \cdot 900t).$$

What sampling rates allow exact reconstruction? Check $f_s = 2000$ and $f_s = 1600$.

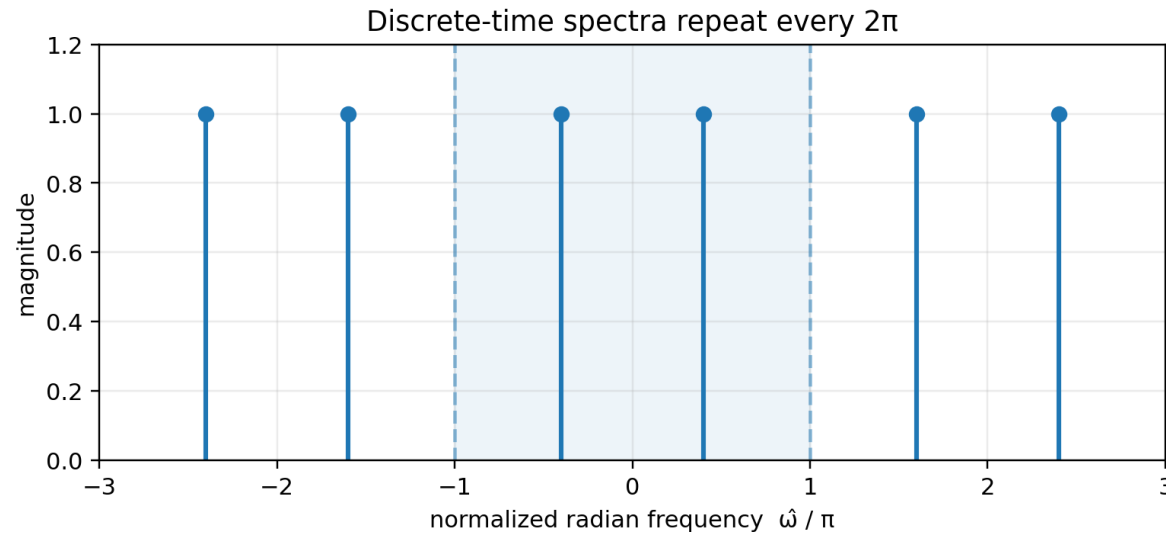
Solution

- 1 The highest non-zero frequency is $f_{\max} = 900$ Hz.
- 2 Sampling theorem: f_s must be greater than $2f_{\max} = 1800$ samples/s.
- 3 $f_s = 2000$ is valid: Nyquist frequency is 1000 Hz, above 900 Hz.
- 4 $f_s = 1600$ is not valid: Nyquist frequency is 800 Hz, below 900 Hz.

Final answer: The strict condition is $f_s > 1800$ samples/s; 2000 works, 1600 aliases the 900-Hz component.

Spectral view of sampling

Discrete-time spectra are periodic



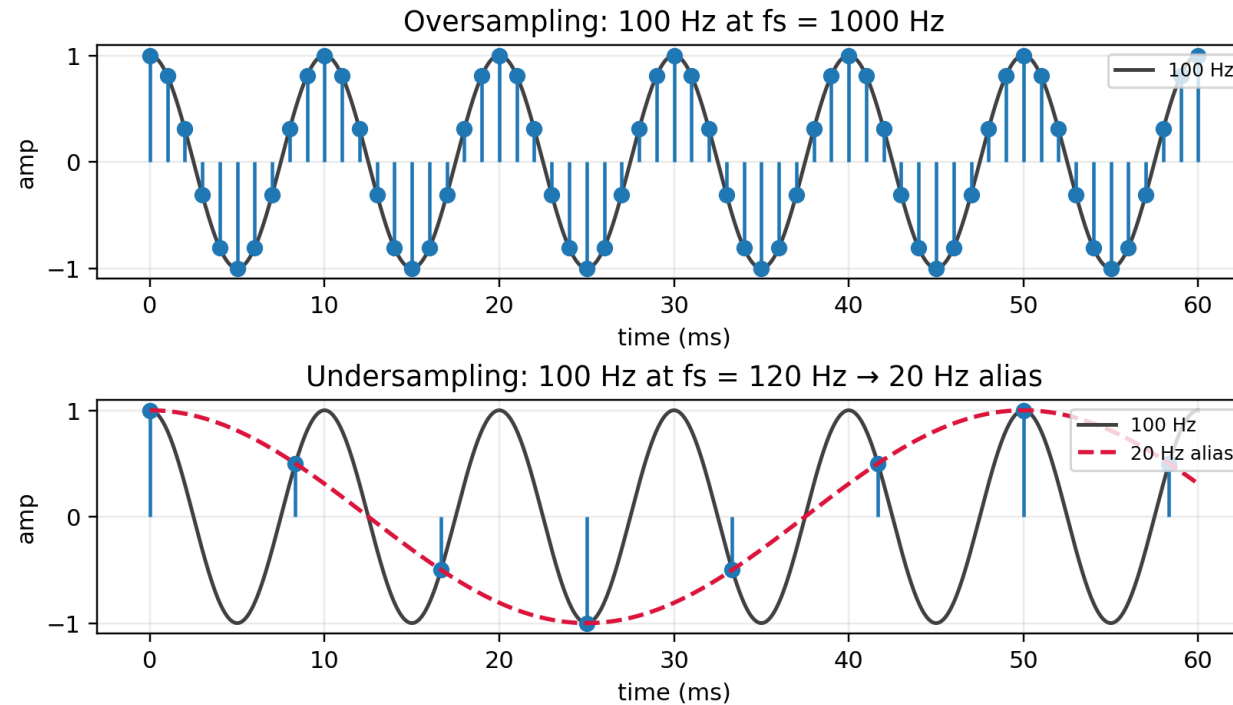
A discrete-time sinusoid has repeated spectral images.
The principal interval is $-\pi \leq \hat{\omega} \leq \pi$.
Reconstruction selects a representative inside this interval.

$$-\pi \leq \hat{\omega} \leq \pi$$

Key idea. Sampling creates periodic copies in normalized frequency.

Oversampling

Sampling comfortably above the Nyquist rate



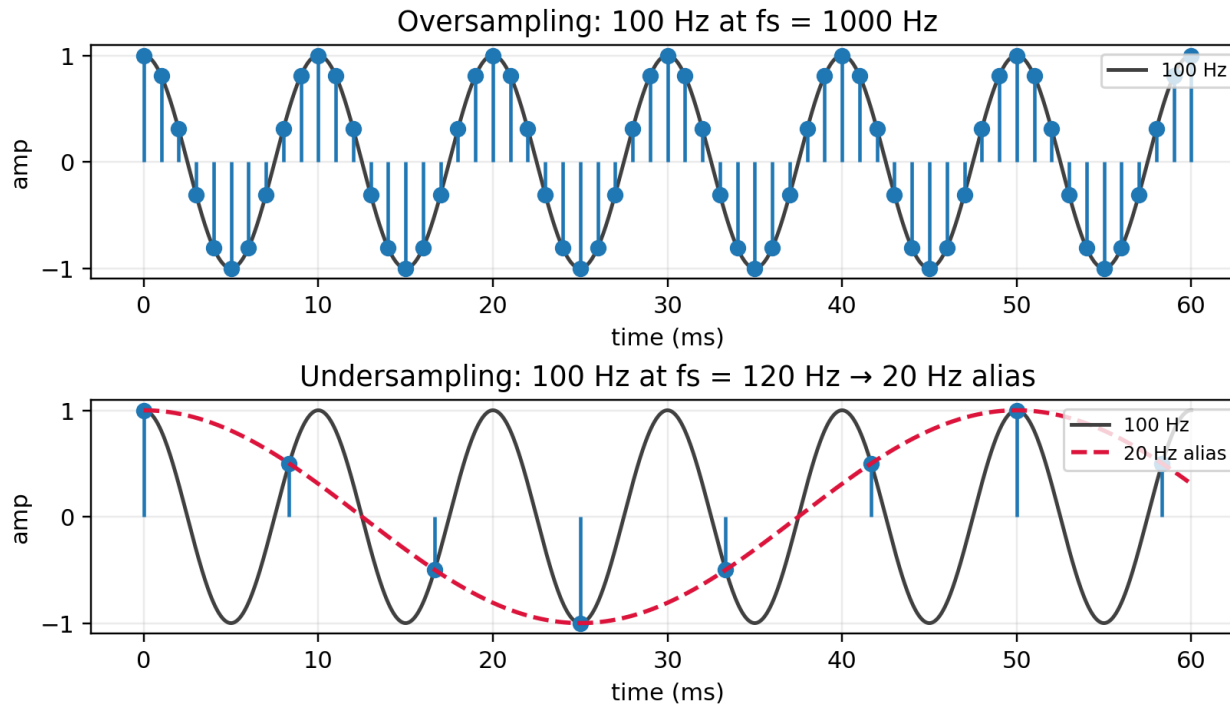
- In practice, we choose f_s well above $2f_{\max}$.
- Oversampling leaves room for anti-aliasing filters.
- The discrete-time sequence retains the intended frequency.

$$\hat{f}_0 = f_0 / f_s, \quad \hat{\omega}_0 = 2\pi\hat{f}_0$$

Key idea. Oversampling makes aliasing less likely and reconstruction easier.

Aliasing caused by undersampling

A high-frequency sinusoid can look slow



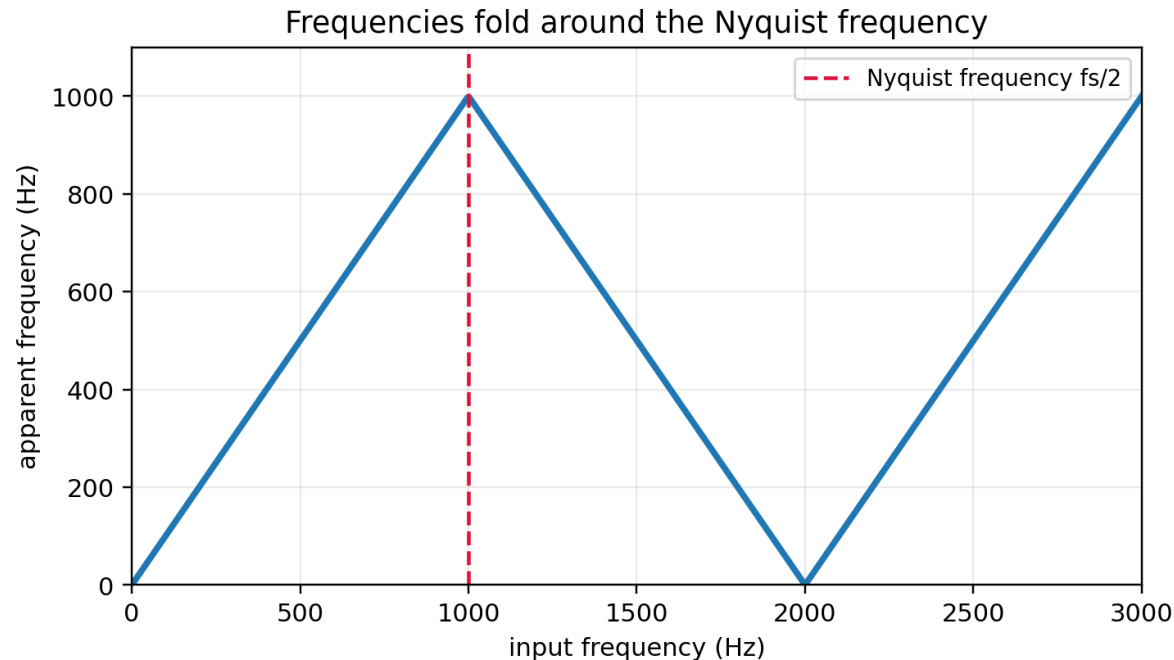
100 Hz sampled at 120 Hz $\rightarrow$ 20 Hz alias

- The original frequency lies above Nyquist: $f_N = 60$ Hz.
- The reconstructed low-frequency sinusoid is not the original.
- Aliasing is irreversible without prior information.

Key idea. Undersampling replaces the original frequency by an alias inside the reconstructible range.

Folding and maximum reconstruction frequency

Frequencies above Nyquist fold back



$$f_{\text{max, recon}} = f_s / 2$$

- The Nyquist frequency is a folding boundary.
- Frequencies above $f_s/2$ appear mirrored into the interval.
- The apparent frequency is the alias closest to zero.

Key idea. The Nyquist interval is the only unambiguous reconstruction interval.

Example 3 — undersampling and folding

A 1000-Hz sinusoid sampled at different rates

Problem

A sinusoid has $f_0=1000$ Hz. For $f_s = 3000$, 2200, 1800, 1250, and 800 samples/s, identify the apparent reconstructed frequency.

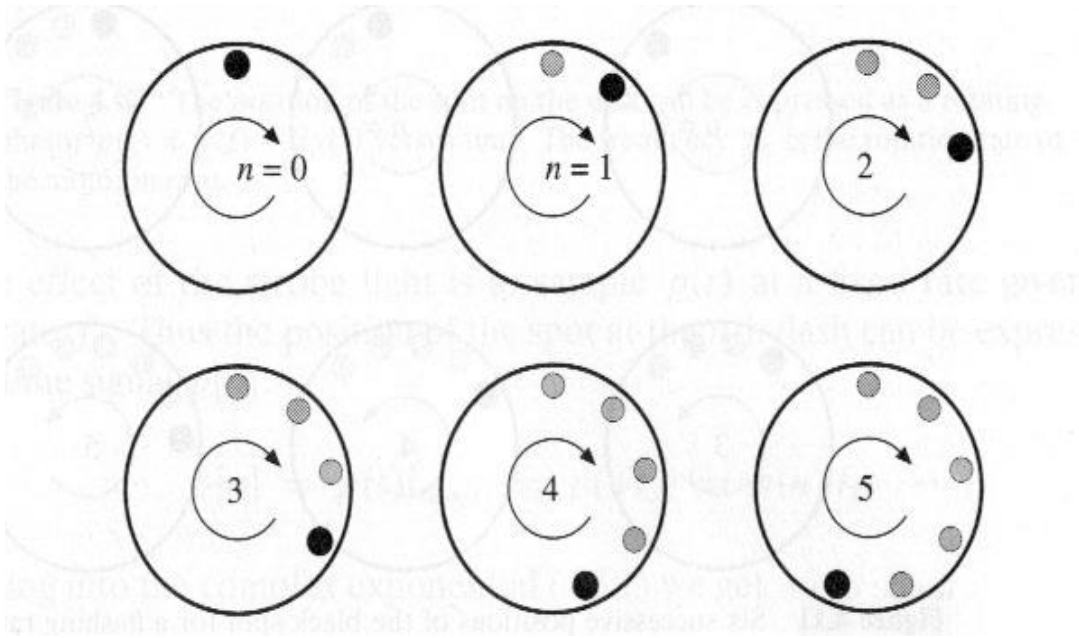
Solution

- 1 Exact reconstruction requires $f_s > 2f_0 = 2000$ samples/s.
- 2 For $f_s=3000$ and 2200, the 1000-Hz component is inside the Nyquist interval.
- 3 For $f_s=1800$: 1000 folds to $|1000-1800| = 800$ Hz.
- 4 For $f_s=1250$ and 800: aliases are $|1000-1250|=250$ Hz and $|1000-800|=200$ Hz.

Final answer: Apparent frequencies: 1000, 1000, 800, 250, and 200 Hz.

Stroboscopic aliasing

Sampling a rotating object with flashes



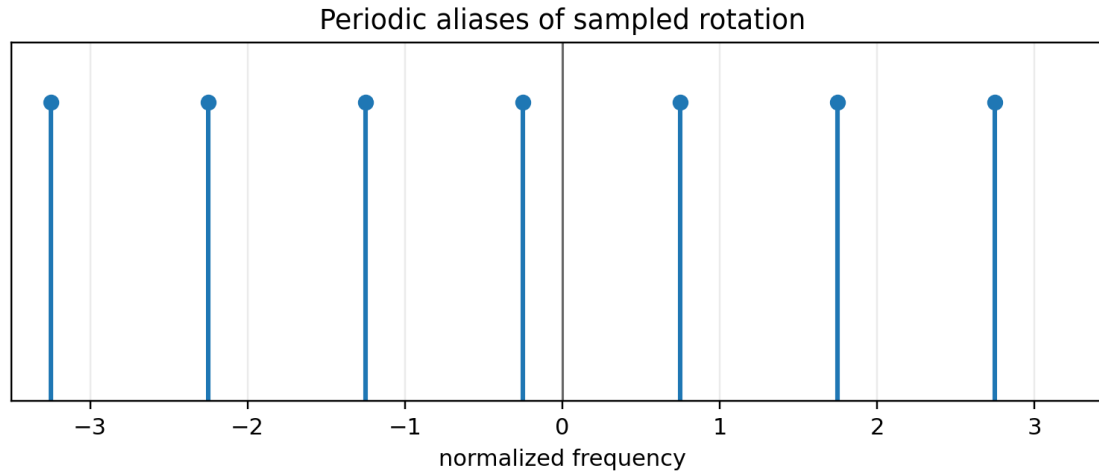
- A strobe light samples the position of a rotating dot.
- If the flash rate matches the rotation rate, the dot appears stationary.
- If the flash rate is slightly different, the dot may appear to rotate slowly or backwards.

apparent frequency = alias closest to 0

Key idea. Aliasing is not only an audio effect; it appears whenever motion is sampled too slowly.

Spectral interpretation of the strobe example

Aliases explain apparent motion



$$f_{obs} = f_m - m f_s$$

- The real rotation has one motor frequency.
- Sampling creates infinitely many aliases.
- The apparent observed motion corresponds to the alias nearest zero.

Key idea. The same aliasing rule explains sounds, sampled sinusoids, and sampled rotation.

Example 4 — stroboscopic aliasing

Aliasing as apparent reverse motion

Problem

A disk rotates clockwise at 750 revolutions/min. It is observed with a strobe. What happens for flash rates 750, 375, and 806 flashes/min?

Solution

- 1 At 750 flashes/min, the disk completes 1 revolution between flashes: it appears stationary.
- 2 At 375 flashes/min, it completes 2 revolutions between flashes: it again appears stationary.
- 3 At 806 flashes/min, the closest alias is $750 - 806 = -56$ revolutions/min.
- 4 The minus sign means the apparent motion is in the opposite direction.

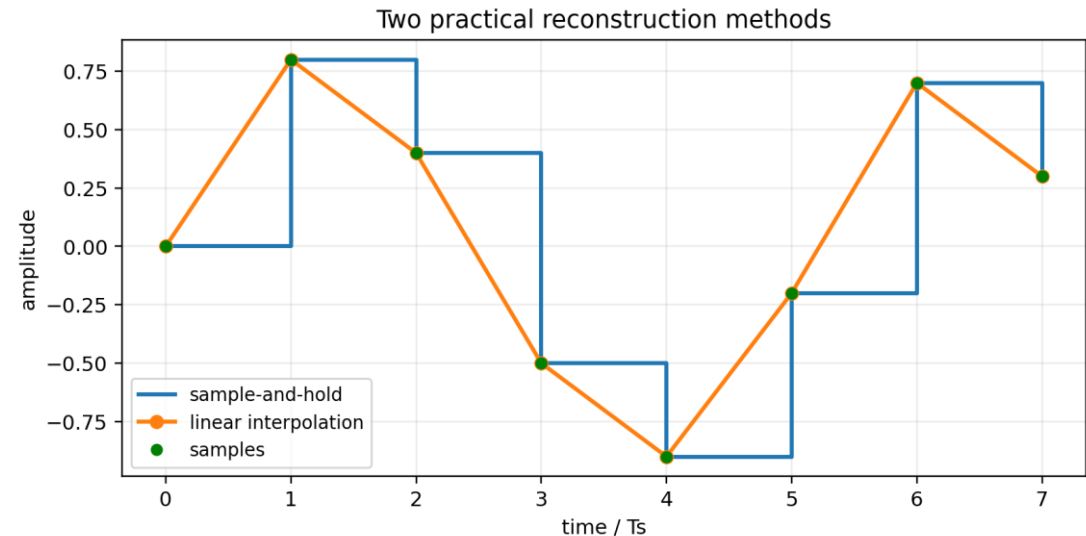
Final answer: The disk appears stationary at 750 and 375 flashes/min, and rotates backwards at 56 rpm for 806 flashes/min.

Discrete-to-continuous conversion

Reconstruction from samples

$$\hat{x}(t) = \sum_n x[n] p(t - nT_s)$$

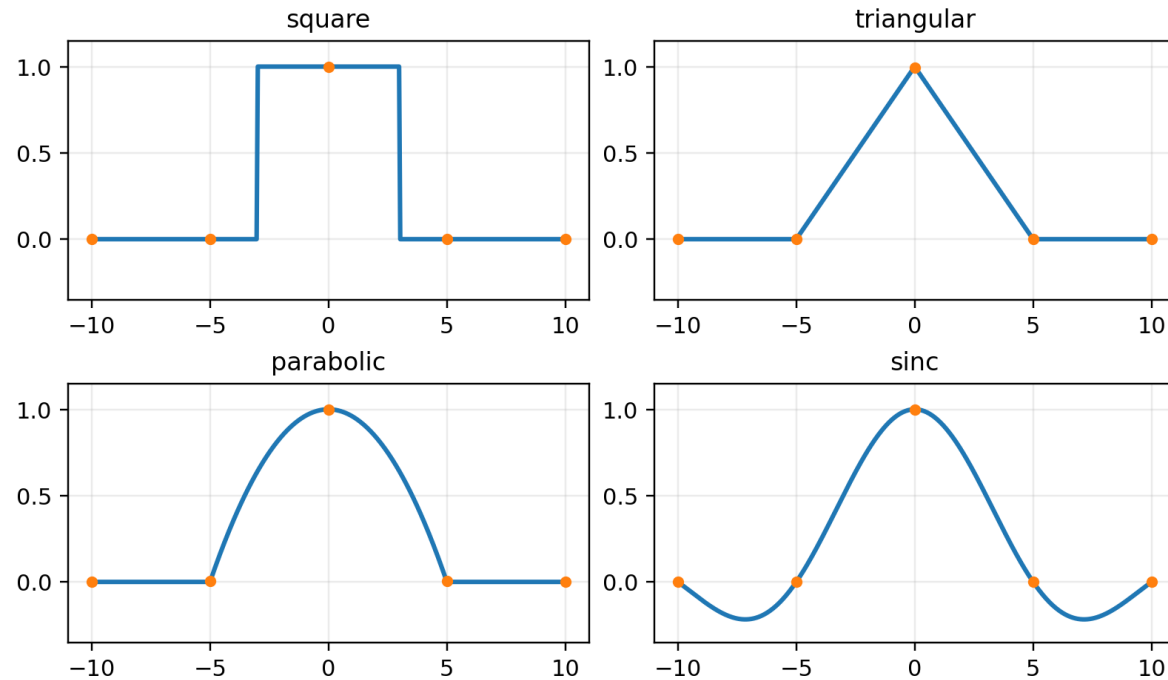
- A D/A converter emits one pulse per sample.
- The interpolation pulse $p(t)$ determines the shape between samples.
- Ideal reconstruction requires a band-limited interpolation pulse.



Key idea. Reconstruction is not just connecting dots; it is a filtering/interpolation operation.

Interpolation pulses

Different pulse shapes give different reconstructions



- Rectangular pulse: zero-order hold / staircase output.
- Triangular pulse: linear interpolation.
- Smoother pulses create smoother reconstructed waveforms.
- The ideal pulse is sinc-shaped.

Key idea. The pulse shape determines how sample values become a continuous-time waveform.

Reconstruction can preserve or distort

It depends on whether the samples are unambiguous

- If $f_0 < f_s/2$, ideal reconstruction recovers the original sinusoid.
- If $f_0 > f_s/2$, reconstruction gives the alias inside the Nyquist interval.
- Anti-aliasing filters remove frequencies that cannot be represented safely.



safe band: $|f| < f_s/2$

Key idea. Good sampling starts before the sampler: remove frequencies above Nyquist.

Notation and takeaways

Minimum language for sampling and aliasing

t continuous-time variable

n sample index

T_s sampling period

f_s sampling rate

f_N Nyquist frequency

$\hat{\omega}_0$ normalized radian frequency

f_0 normalized cyclic frequency

$\hat{x}(t)$ reconstructed signal

- Sampling converts continuous time to an indexed sequence.
- Aliasing happens when different analogue frequencies produce the same samples.
- The sampling theorem gives the condition for exact reconstruction.
- Interpolation determines the reconstructed waveform between samples.